

Lab 01 – Week 01

Identifying Super Keys, Candidate Keys and Primary key from Database Tables

Problem Statement:

Suppose, you are hired as a junior database developer in an XYZ software house. The software house gets a new project to develop the database of an ABC university. As a part of database development team, you are assigned the following tasks to perform.

Task 1:

Create a table named “Student” which should have the following attributes:

Roll Number, Student Name, Father Name, CNIC#, Mobile#, Degree Program, CGPA.

Roll Number	Student Name	Father Name	CNIC#	Mobile#	Degree Program	CGPA

Task 2:

Identify Super Keys, and Candidate Keys for the given table “Student”.

Super Key: {Roll Number}, {CNIC#}, {Roll Number, Student Name}, {CNIC#, Student Name}, {Roll Number, Student Name, Father Name, CNIC#, Mobile#, Degree Program, CGPA}

Candidate Keys: {Roll Number}, {CNIC#}, {Mobile#}

Task 3:

Create another table named “Employee” with following attributes:

Emp ID, Name, CNIC, Salary, Designation, Email ID.

Emp ID	Name	CNIC	Salary	Designation,	Email ID

Task 4:

Identify Candidate Keys and Primary Key for the given table “Employee”.

Candidate Keys: {Emp ID}, {CNIC}, {Email ID}

Primary Key: {Emp ID}

Good Luck!

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Lab 02 – Week 02

Identifying Surrogated Keys and Foreign key from Database Tables

Problem Statement:

Suppose, you are hired as a junior database developer in an Abbot's solution. The software house gets a new project to develop the database for a new university. As a part of database development team, you are assigned the following tasks to perform.

Task 1:

Create a table named "Student" which should have the following attributes:

Roll Number, Student Name, Father Name, CNIC#, Mobile#, Degree Program, CGPA.

Roll Number	Student Name	Father Name	CNIC#	Mobile#	Degree Program	CGPA

Task 2:

Identify Surrogated Keys for the given table "Student".

Surrogated Keys: { Application Number }

Task 3:

Create another table named "Employee" with following attributes:

Emp ID, Name, CNIC, Salary, Designation, Email ID.

Emp ID	Name	CNIC	Salary	Designation	Email ID

Create another table named "Department" with following attributes:

Dpt ID, Dpt Name, Emp ID, Tel No

Dpt ID	Dpt Name	Emp ID	Tel No

Task 4:

Identify Primary key and Foreign Key for the given tables “Employee” and “Department”.

Employee Table:

Primary key: {Emp ID}

Foreign Key: {}

Department Table:

Primary key: {Dpt ID}

Foreign Key: {Emp ID}

Good Luck!

Lab 03 – Week 03

Problem Statement:

The ABC Company aims to develop a project management software which enables its users to manage and track projects, deadlines, schedules, and other workflow processes.

Each user has to perform a specific role. User can create tasks along with discussion threads. Users can post messages in Discussion Threads. User can also able to edit tasks and set deadlines. A complete progress can be viewed for the specific task.

You're required to perform the following tasks according to above scenario.

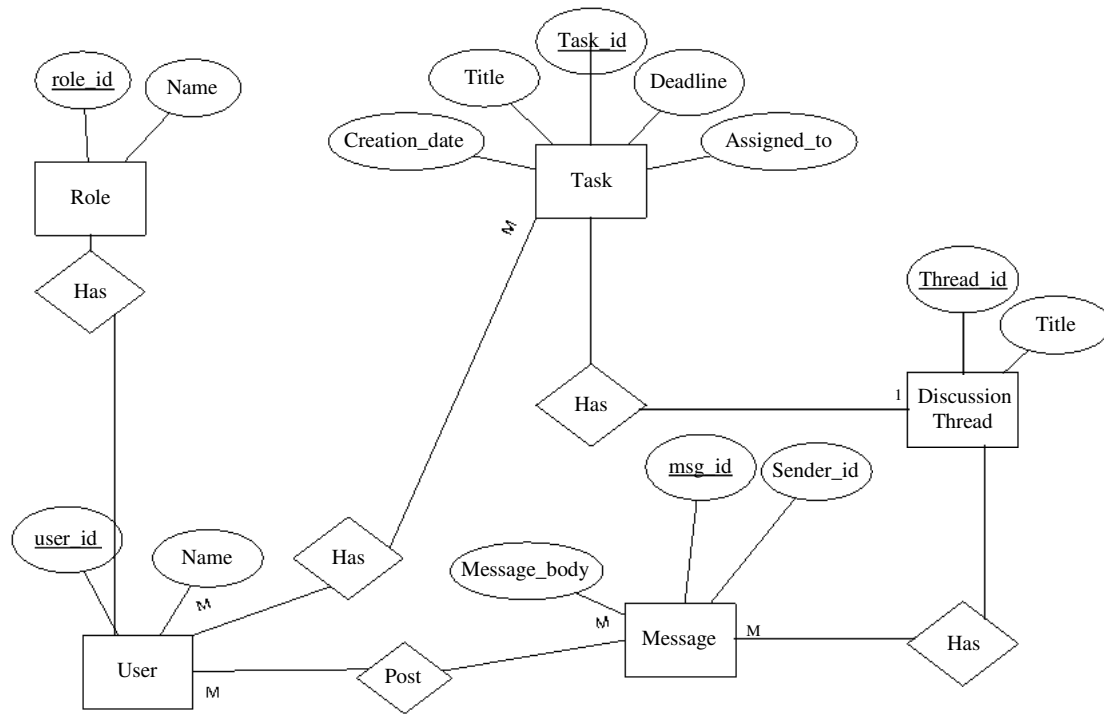
Task 1

Identify Entities, their attributes, relationships among entities in above scenario

<u>Entities</u>	<u>Attributes</u>
Role	role_id, name
User	user_id, name
Task	Task_id, title, creation_date, deadline, assigned_to
Discussion Thread	Thread_id, title
Messages	msg_id, message_body, sender_id

Task 2

Draw Entity Relationship Diagram (ERD) for it.



Lab 04: Installation of Oracle 11g

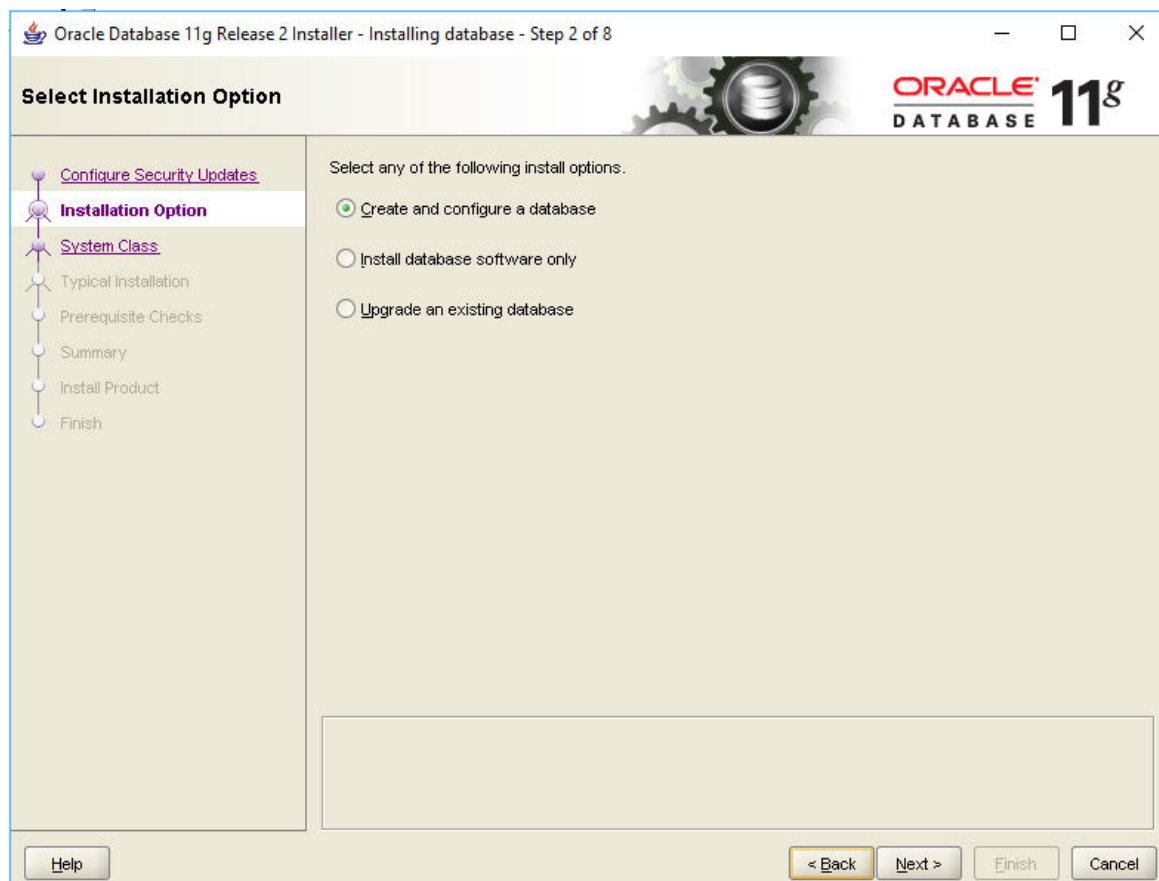
Task 1: You are required to install Oracle 11g in your system.

Installation Steps:

Before Installing Oracle, you must install complete OS

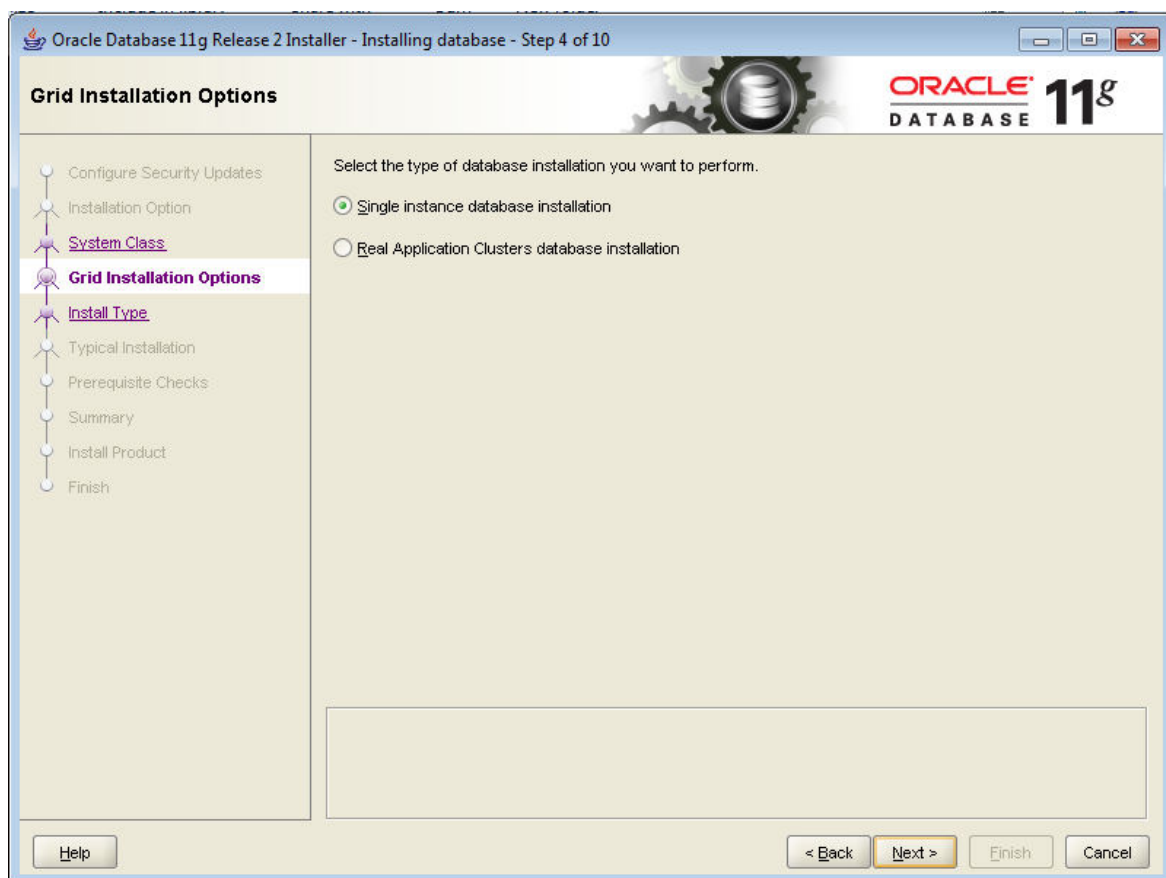
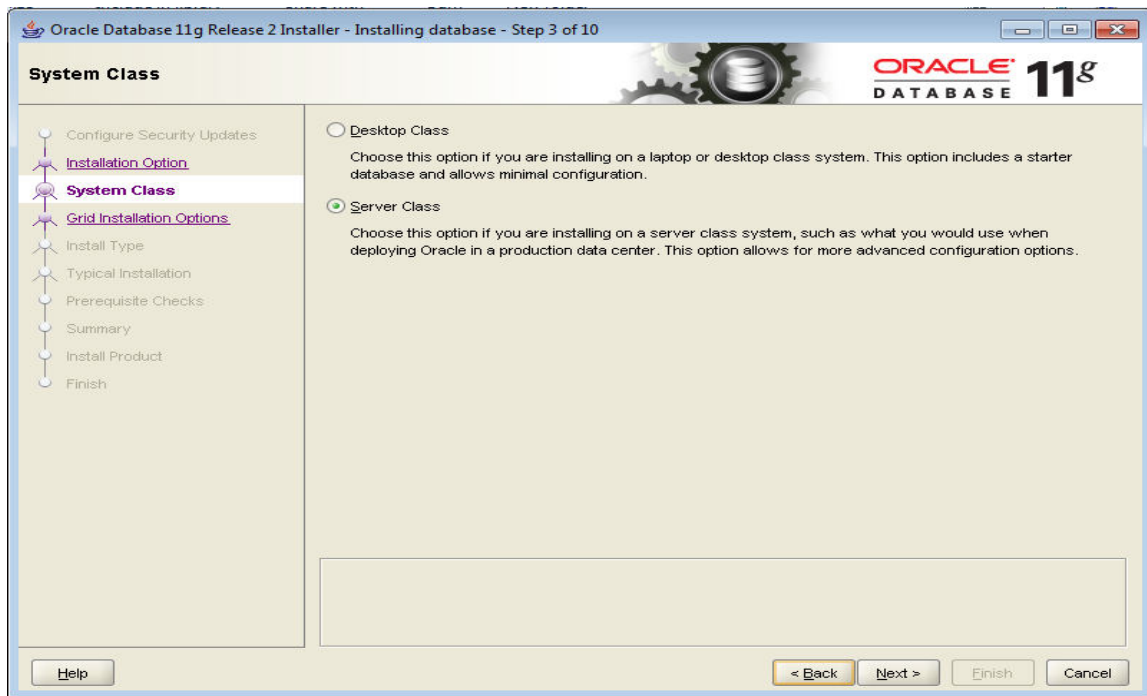
windows with all patches and activations. You may get this pop up in case if you have oracle previous

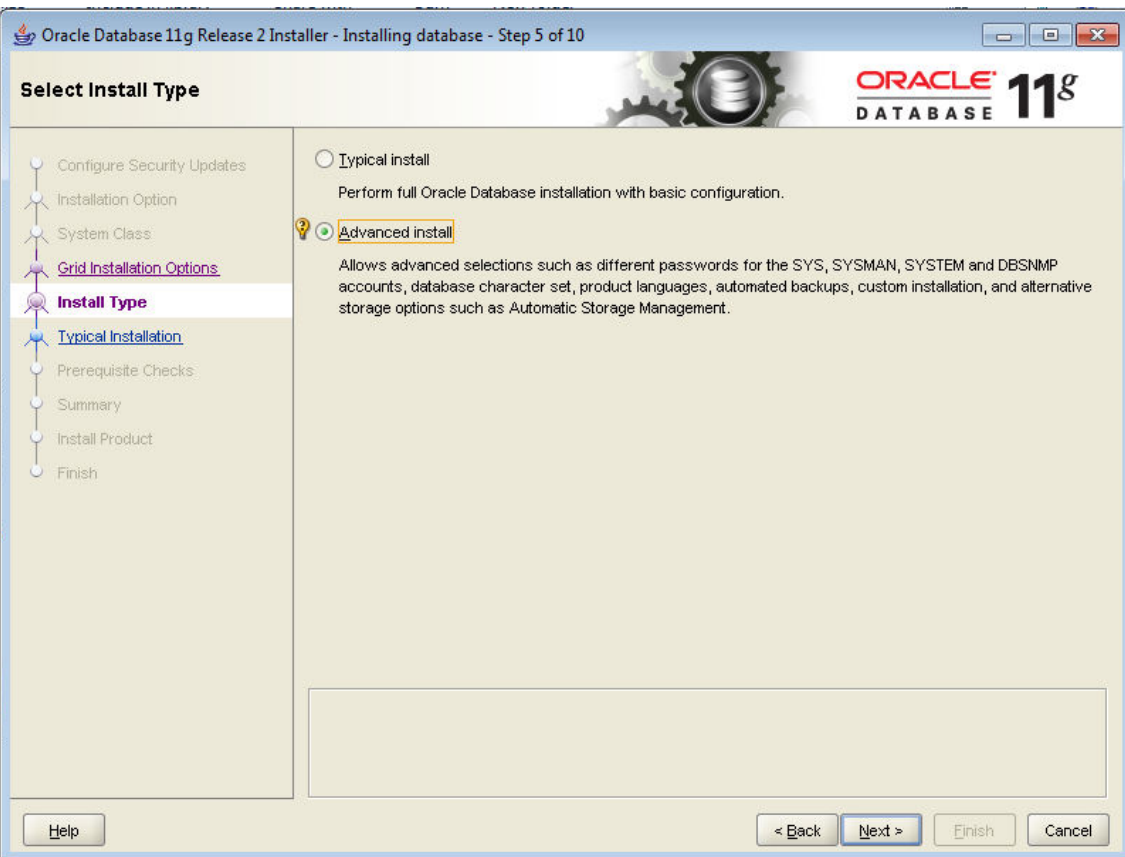
Installation, other proceed with next choice.

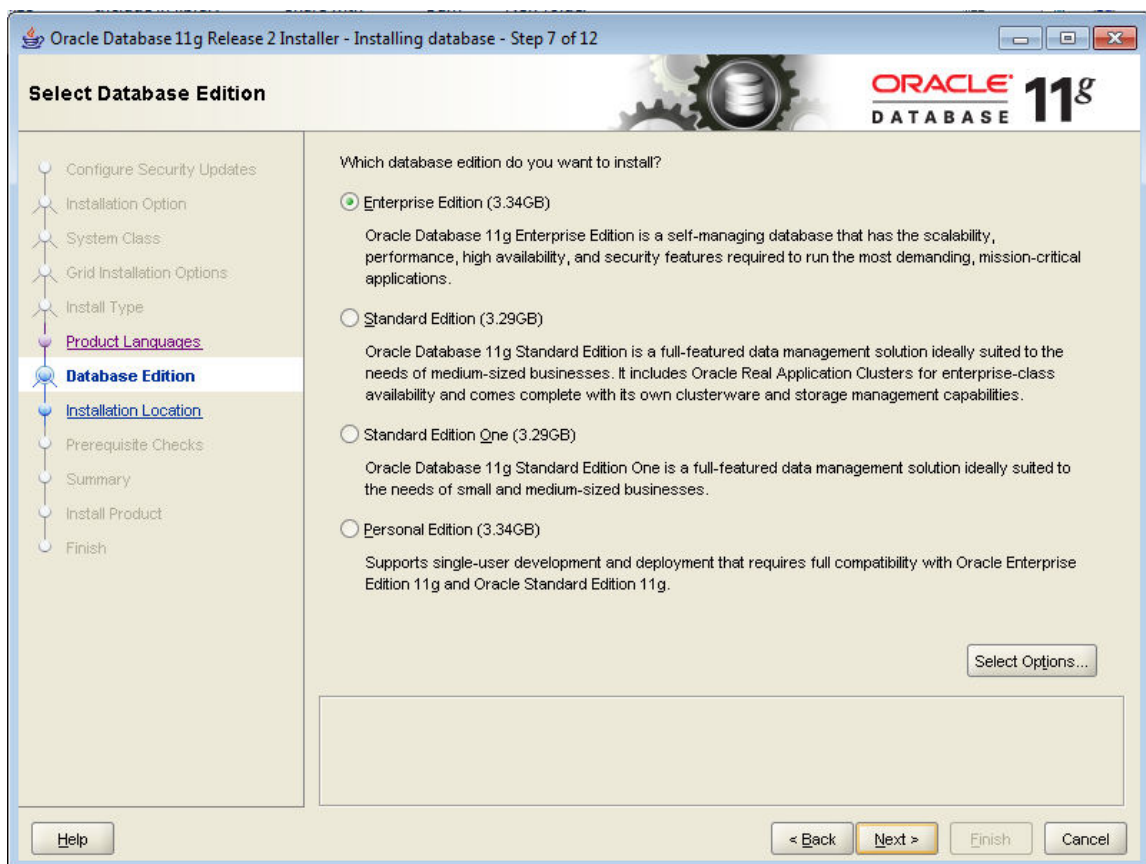
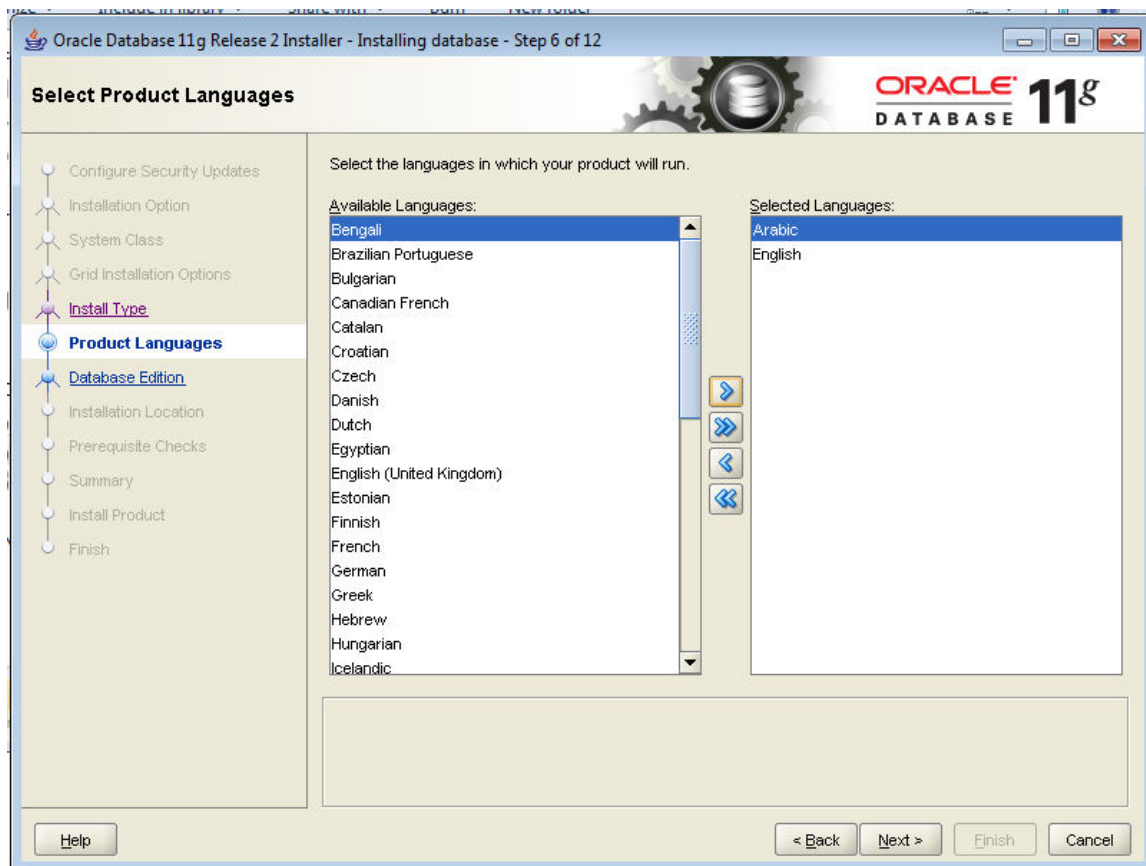


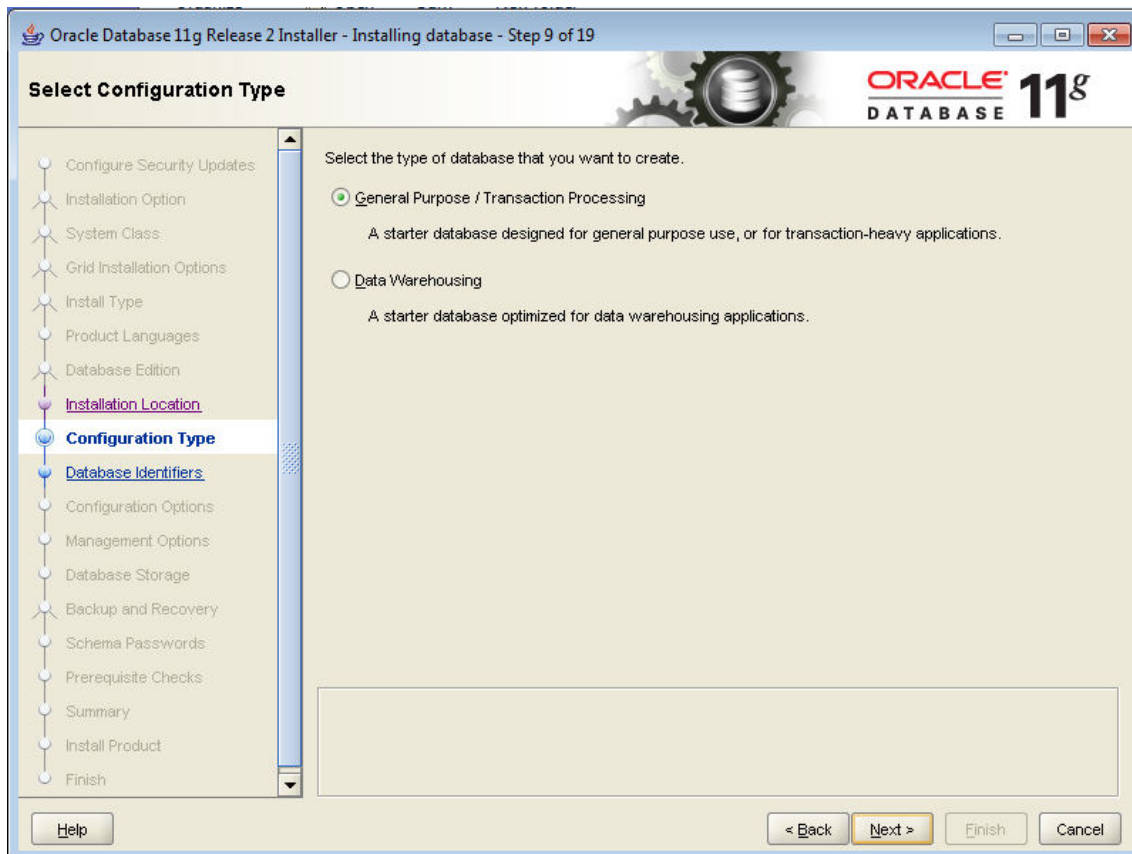
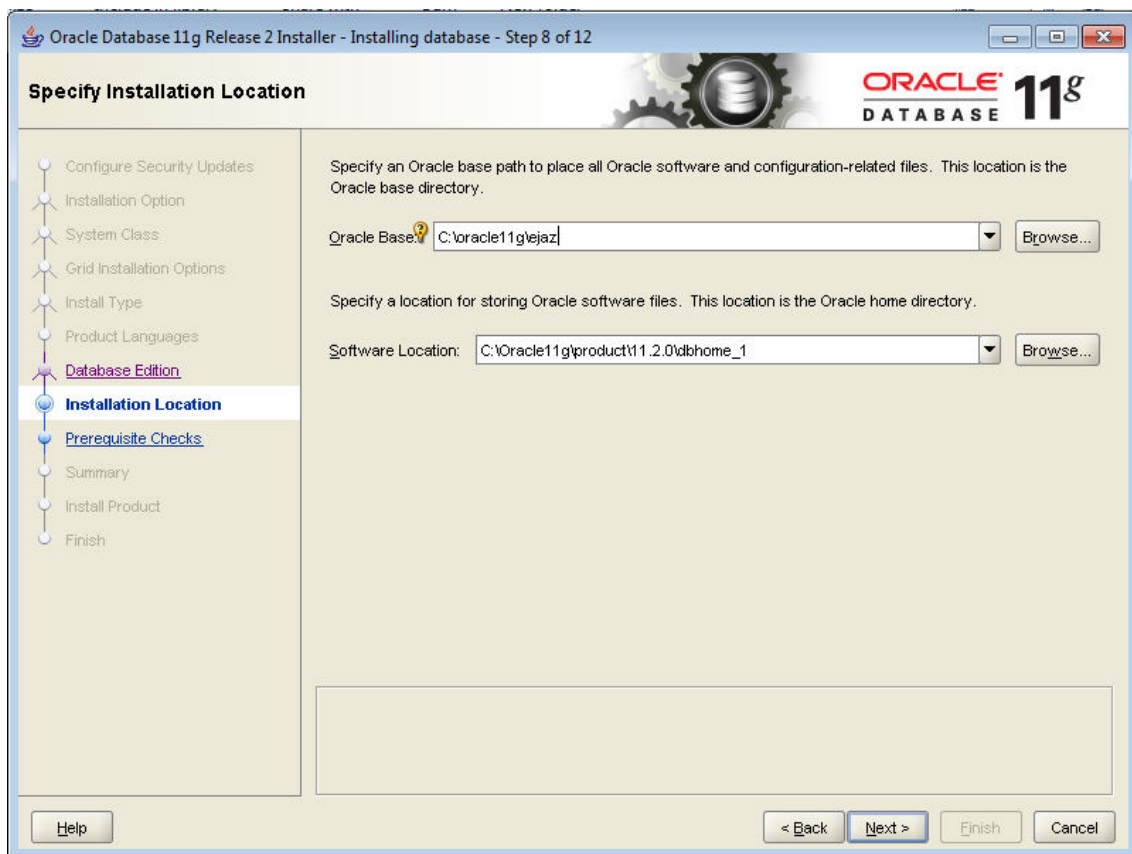
Before Installing Oracle, you must install complete OS windows with all patches and activations

Run SETUP.EXE to start Oracle Universal Installer.









Oracle Database 11g Release 2 Installer - Installing database - Step 10 of 19

Specify Database Identifiers

Configure Security Updates

Installation Option

System Class

Grid Installation Options

Install Type

Product Languages

Database Edition

Installation Location

Configuration Type

Database Identifiers

Configuration Options

Management Options

Database Storage

Backup and Recovery

Schema Passwords

Prerequisite Checks

Summary

Install Product

Finish

An Oracle database is uniquely identified by a Global Database Name, typically of the form "name.domain".

Global database name:

A database is referenced by at least one Oracle instance that is uniquely identified from any other instance on this computer by an Oracle Service Identifier (SID).

Oracle Service Identifier (SID):

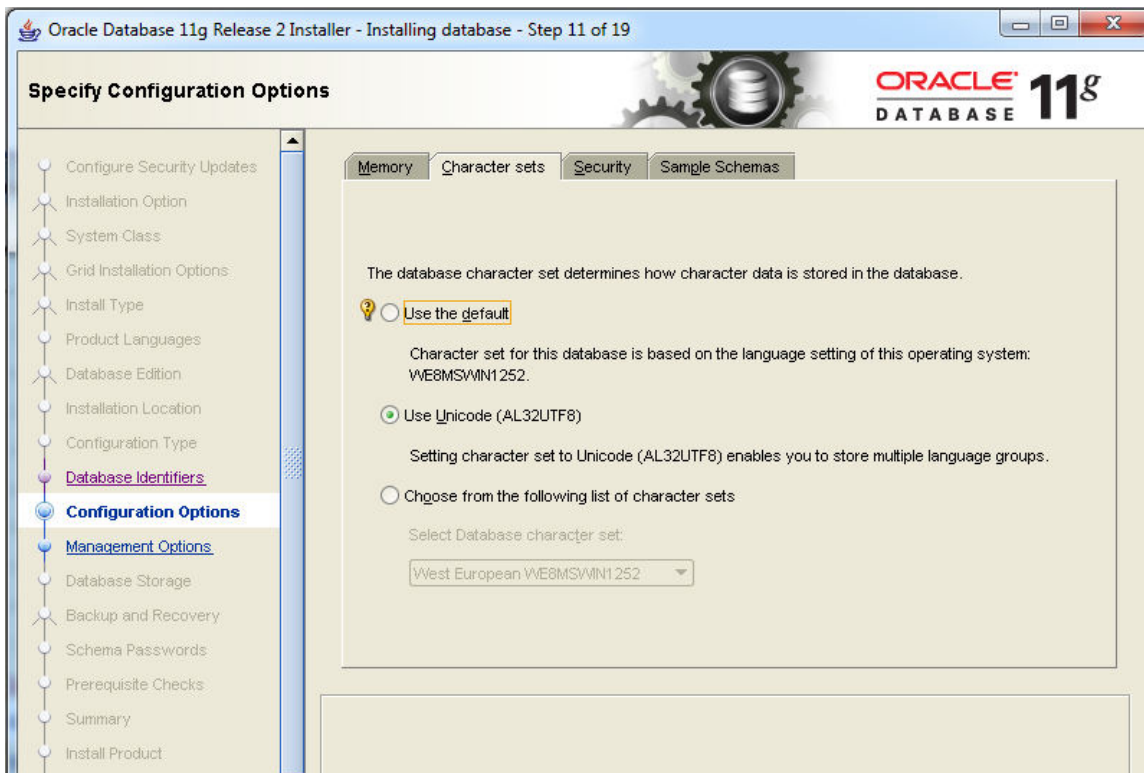
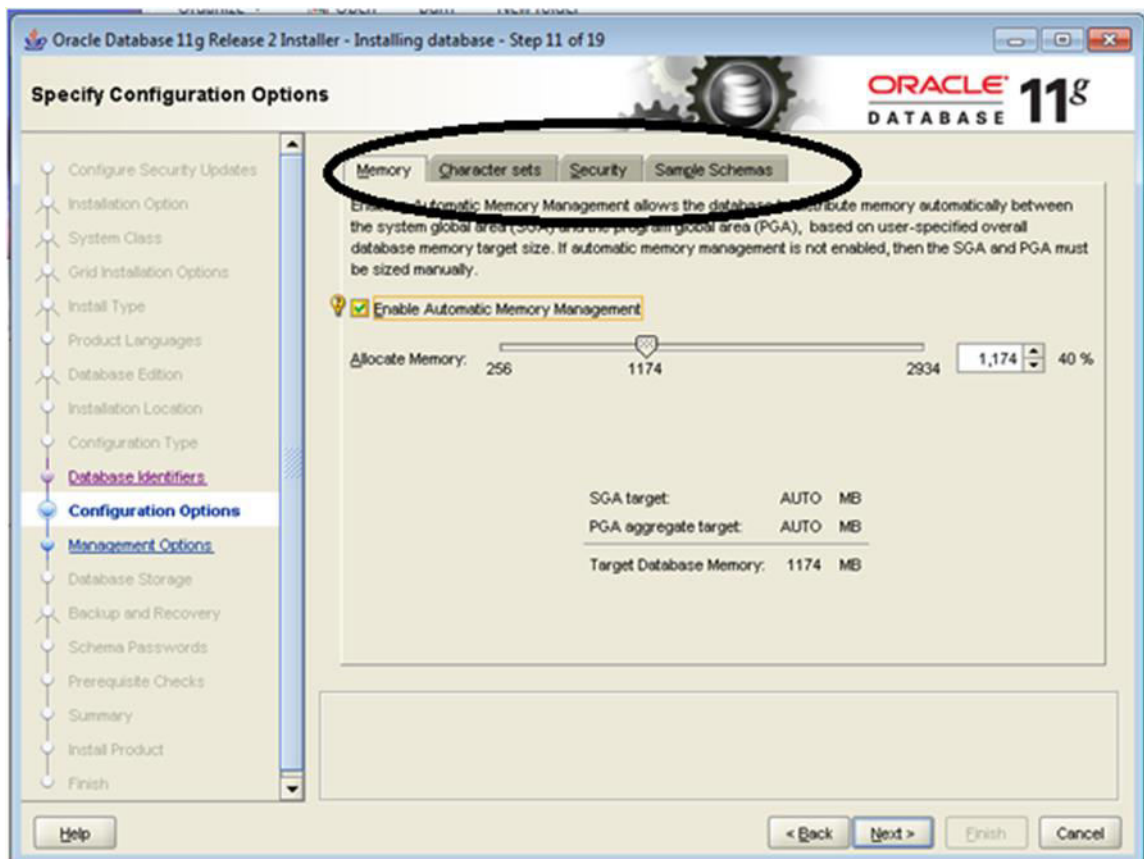
Help

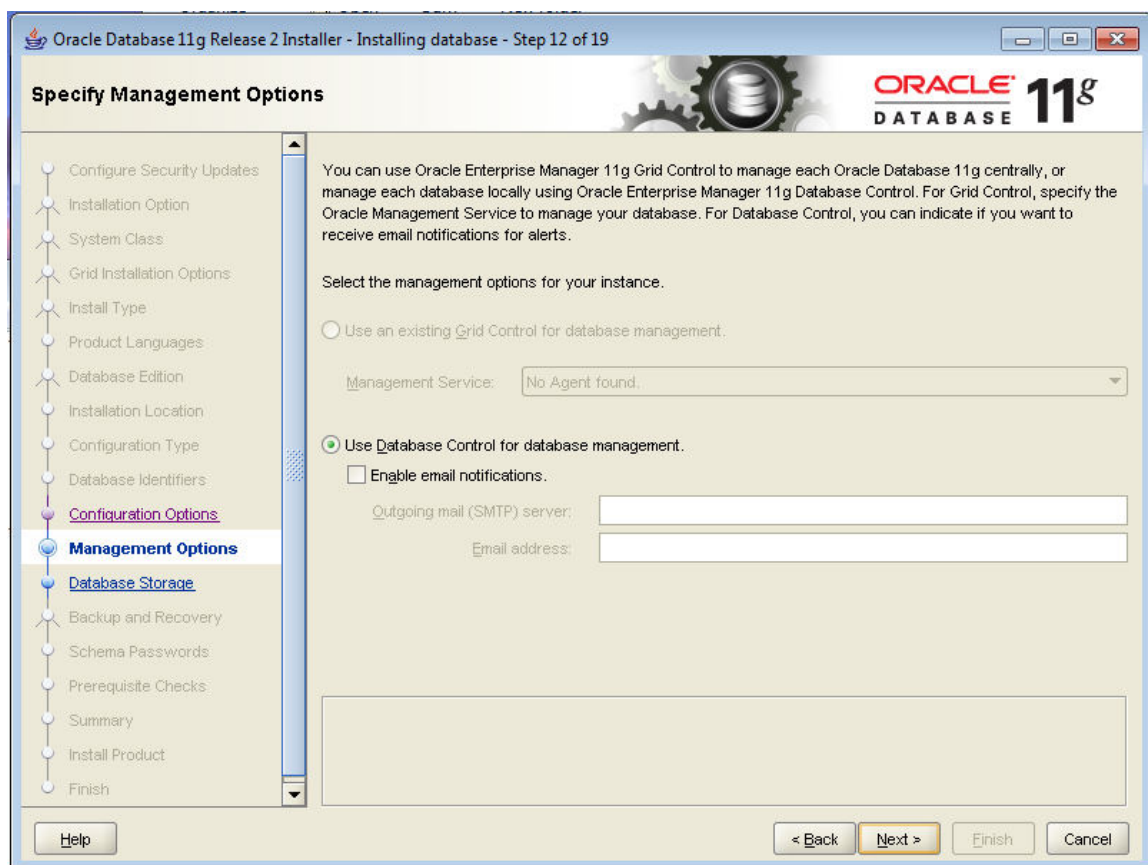
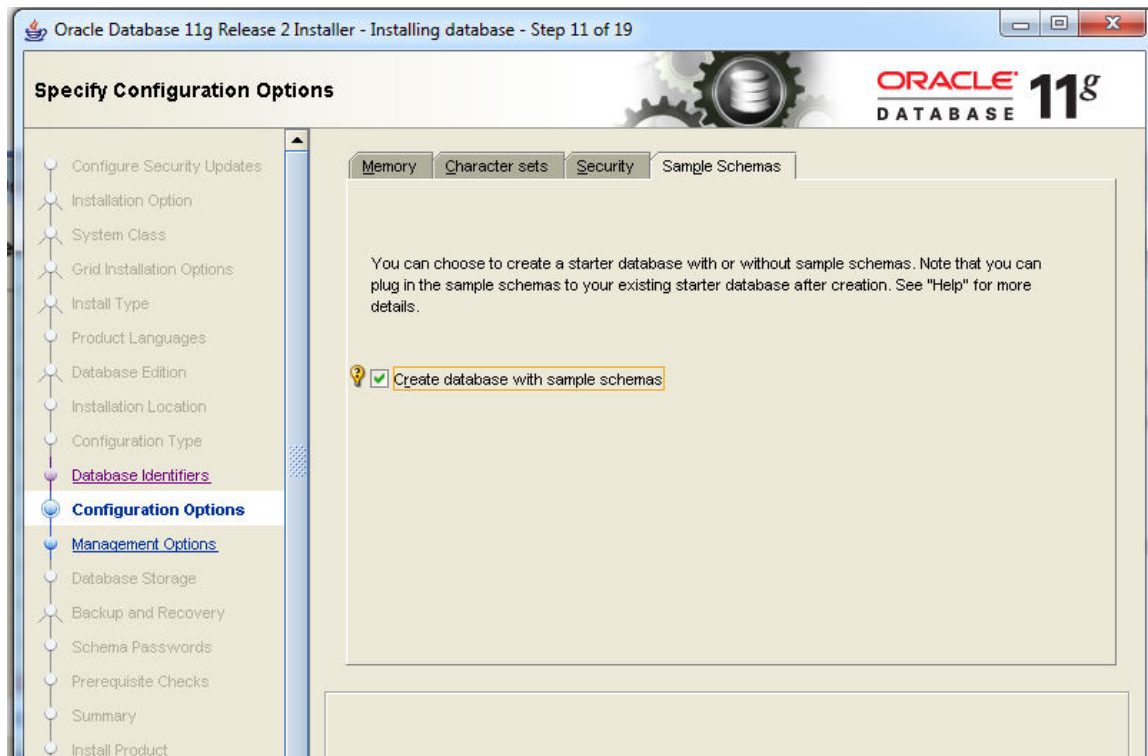
< Back

Next >

Finish

Cancel





Oracle Database 11g Release 2 Installer - Installing database - Step 13 of 19

Specify Database Storage Options

File System

Use a file system for database storage. For optimal database organization and performance, Oracle recommends that you install data files and the Oracle database software on different disks. If you intend to use Automatic Storage Manager File System (ASMFS) as the storage option, then select Automatic Storage Management.

Specify database file location:

☐ Automatic Storage Management

Automatic Storage Management (ASM) simplifies database storage administration, and places database files for optimal I/O performance. Select this option if you intend to use either ASM, or Automatic Storage Management File System.

Specify password of ASMSNMP user:

Oracle Database 11g Release 2 Installer - Installing database - Step 14 of 19

Specify Recovery Options

Select to enable or disable automated backups for your database. If enabled, then the Backup Job uses the specified recovery area storage.

☒ Do not enable automated backups

☐ Enable automated backups

Recovery area storage

☒ File System

Recovery Area location:

☐ Automatic Storage Management

Backup Job Operating System credentials

Username: Password:

Oracle Database 11g Release 2 Installer - Installing database - Step 15 of 19

Specify Schema Passwords

The starter database contains pre-loaded schemas, most of which have passwords that are expired and locked at the end of installation. After installation is complete, you must unlock and set new passwords for those accounts you want to use. Schemas used for database management and postinstallation functions are left unlocked, and passwords for these accounts will not expire. Specify the passwords for these accounts.

☐ Use different passwords for these accounts

	Password	Confirm password
SYS	*****	*****
SYSTEM	*****	*****
SYSTEM	*****	*****
DBSNMP	*****	*****

☒ Use the same password for all accounts

Password: ***** Confirm password? *****

Messages:

⚠ Password:[INS-30011] The ADMIN password entered does not conform to the Oracle recommended standards.

Help < Back Next > Finish Cancel

Oracle Database 11g Release 2 Installer - Installing database - Step 18 of 19

Install Product

Progress

8%

Loading products. Please wait.

Status

Oracle Database installation	In Progress
• Prepare	Succeeded
• Copy files	Pending
• Setup files	Pending
Oracle Database configuration	Pending

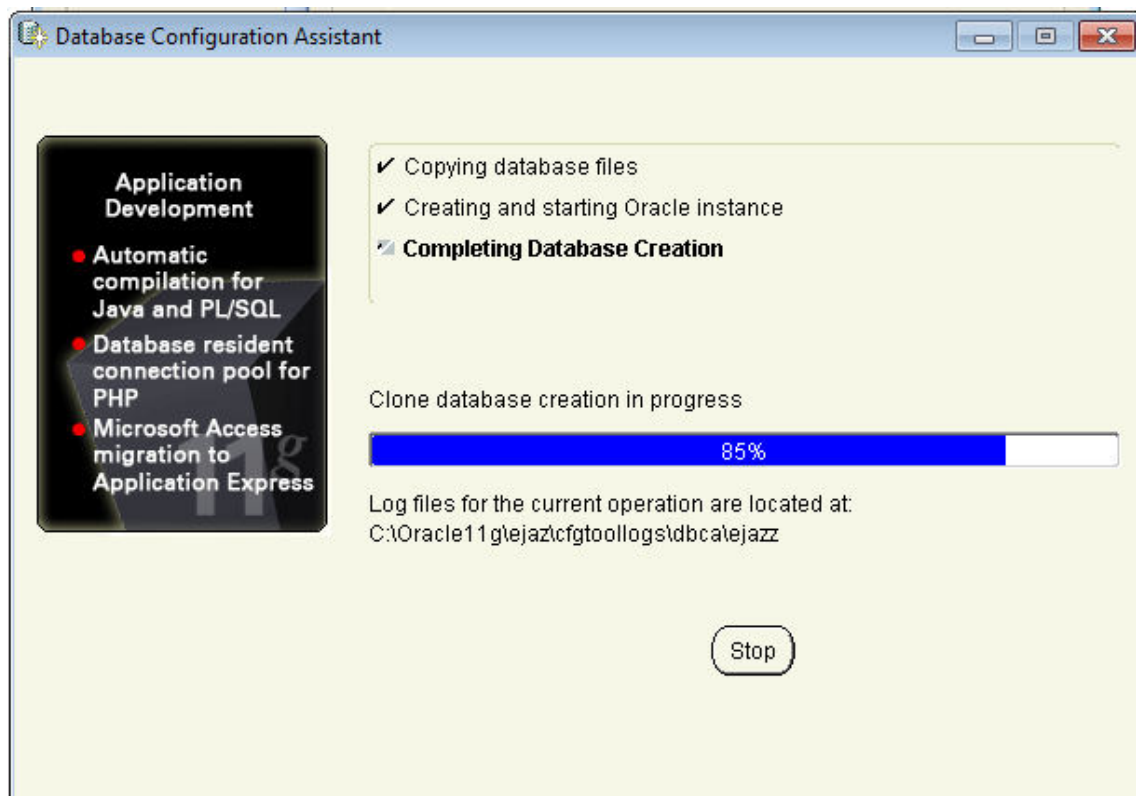
Details Retry Skip

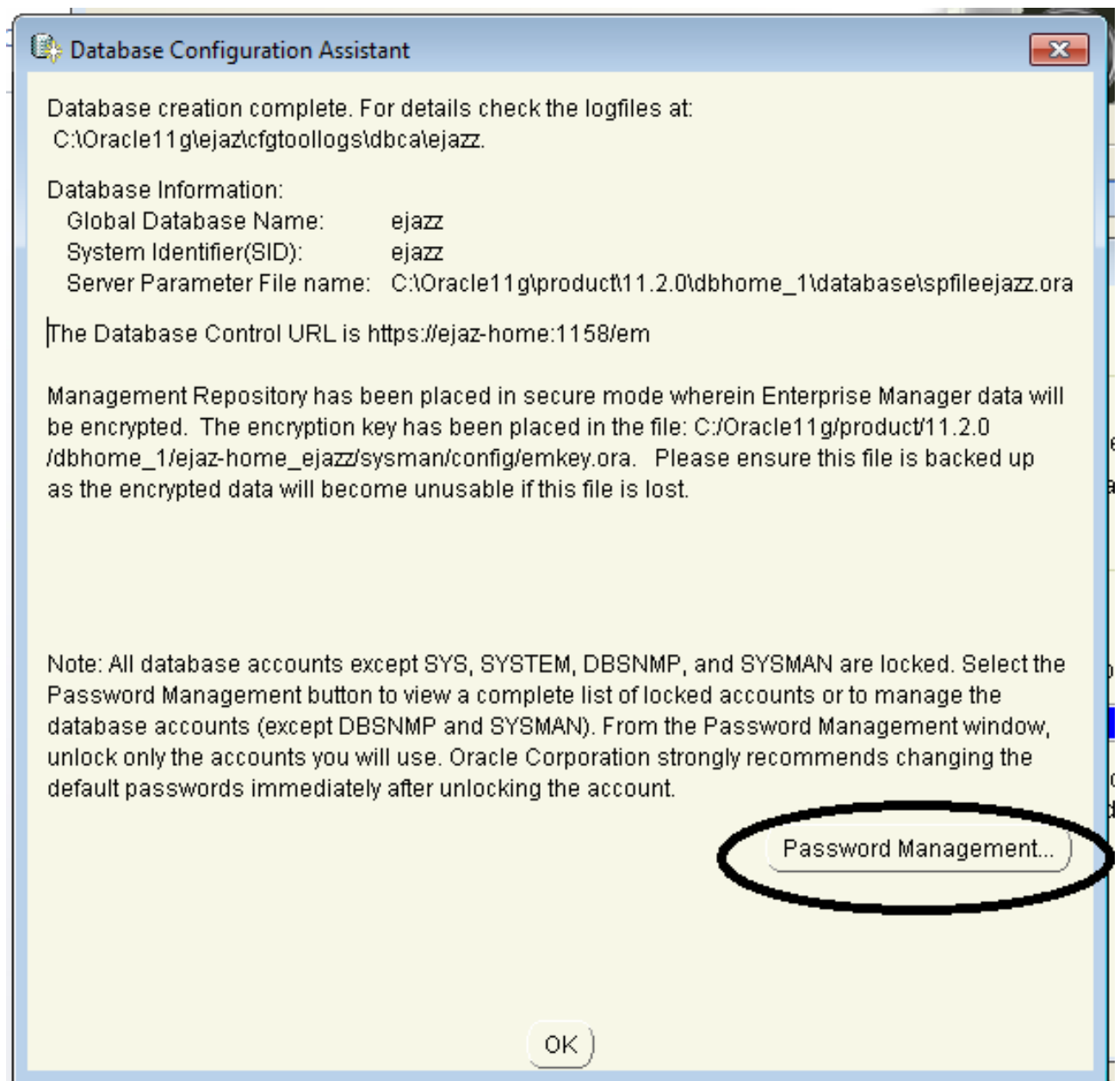
Oracle Database 11g Consolidate Compress Control

Help < Back Next > Finish Cancel

In case if any process failed, press Retry button.

If NetConfig error displays then press continue





Login with Oracle Database Manager

ORACLE Enterprise Manager 11g
Database Control

Login

* User Name
* Password
Connect As

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The screenshot shows a Windows desktop environment. On the left, the Start menu is open, displaying a list of applications and folders. The 'SQL Plus' application is highlighted with a red circle. The main window of the desktop is a terminal window titled 'SQL Plus'. The terminal displays the following text:

```
SQL*Plus: Release 11.2.0.1.0 Production on Wed Sep 12 09:23:55 2012
Copyright (c) 1982, 2010, Oracle. All rights reserved.
Enter user-name: scott/tiger@ejazz
ERROR:
ORA-28001: the password has expired

Changing password for scott
New password:
Retype new password:
Password changed

Connected to:
Oracle Database 11g Enterprise Edition Release 11.2.0.1.0 - 64bit Production
With the Partitioning, OLAP, Data Mining and Real Application Testing options

SQL> select * from tab;

  TNAME          TABTYPE  CLUSTERID
-----
BONUS           TABLE
DEPT             TABLE
EMP             TABLE
SALGRADE        TABLE

SQL>
```

Lab 05: Redo Log Files

Task 1:

Write an SQL statement to create a new group of redo log files. The redo log files names should be respectively log1b.rdo, logbc.rdo and their size should be 100 MB.

```
ALTER DATABASE  
ADD LOGFILE  
(/oracle/dbs/log1b.rdo',  
'/oracle/dbs/log2b.rdo')  
SIZE 100M;
```

Task 2:

Write an SQL statement to create a new group of redo log files with group number. The redo log files names should be respectively log1c.rdo, log2c.rdo. Their size, BLOCK SIZE and group number should be 100 MB, 512, 10 respectively.

```
ALTER DATABASE  
ADD LOGFILE GROUP 10 ('/oracle/dbs/log1c.rdo', '/oracle/dbs/log2c.rdo')  
SIZE 100M BLOCKSIZE 512;
```

Task 3:

Write a SQL following statement that add a new redo log member (log2b.rdo) to redo log group number 3.

```
ALTER DATABASE  
ADD LOGFILE MEMBER '/oracle/dbs/log2b.rdo' TO GROUP 3;
```

Task4:

Write an SQL statement to drop the redo log group number 2?

```
ALTER DATABASE DROP LOGFILE GROUP 2;
```

Lab 06-Week 06

Topic: Tablespaces

Task 1:

Write an SQL statement to create a new tablespace named **NEWTB** and its size should be **50 MB**.

```
CREATE TABLESPACE NEWTB  
datafile 'C:\Oracle11g\oradata\wrk01.dbf'  
size 50M online;
```

Task 2:

Write an SQL statement to create locally-Managed Tablespace named **user1** and its size should be **10 MB**?

```
CREATE TABLESPACE user1  
datafile 'C:\Oracle11g\oradata\usr01.dbf'  
size 10M  
EXTENT MANAGEMENT LOCAL UNIFORM SIZE 128K;
```

Task 3:

Write a SQL statement to create the undo tablespace name **undo10** and its size should be **20MB**.

```
CREATE UNDO TABLESPACE undo10  
datafile 'C:\Oracle11g\oradata\undo01.dbf'  
size 20M;
```

Lab 07 - Week 07

Topic: Creating and Alter table

Task 1:

Write SQL command to create the **Employee** table having the following columns and their constraints. You are required to choose the appropriate data type for each column by your own.

Column Name	Data Type	Constraint
EMP_No	-	NOT NULL
EMP_Name	-	NOT NULL
EMP_Age	-	DEFAULT 0
EMP_City	-	
Salary	-	NOT NULL

```
CREATE TABLE Employee (  
EMP_No          INT NOT NULL,  
EMP_Name        Varchar2(20) NOT NULL,  
EMP_Age         Number(5) DEFAULT 0,  
EMP_City        Varchar2(40)  
  
Salary          Number(6) NOT NULL  
);
```

Task 2:

Consider the following Employee table:

EMP_No	EMP_Name	EMP_Age	EMP_City	Salary

You are required to write PL/SQL code for an ORACLE transaction on employee table (given above) containing the following SQL commands.

- INSERT (ADD) a new record having EMP_NO as “006” , EMP_Name as “ALI” , EMP_Age as “30” , EMP_City as “LHR” and Salary as “55000”.
- Delete “EMP_City” column from employee relation.

Solution:

```
DECLARE
```

```
BEGIN
```

```
INSERT INTO Employee (EMP_NO, EMP_Name, EMP_Age, EMP_City, Salary)
VALUES ('006', 'ALI', '30', 'LHR', '55000');
```

```
ALTER TABLE Employee
```

```
Drop column 'EMP_City';
```

```
COMMIT;
```

```
EXCEPTION
```

```
WHEN OTHERS THEN
```

```
ROLLBACK;
```

```
END;
```

```
/
```

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Lab 08 - Week 08

Topic: Create & Alter Index

Task 1:

Write PL/SQL command to create an Index on the name column of give table of employee?

EMP_No	EMP_Name	EMP_Age	EMP_City	Salary

```
CREATE INDEX Emp_idx ON Employee (EMP_Name);
```

Task 2:

Write an PL/SQL command to rename the index called *EMP_idx* to *EMP_index_name*.

```
ALTER INDEX EMP_idx RENAME TO EMP_index_name;
```

Task 3:

Write SQL command to create an Index named *ENAME_index* on the *EMP_Name* column of above given table of employee. Tablespace should be used “users” and storage size is initial 20k and next is 20K.

```
CREATE INDEX ENAME_index ON Employee(EMP_Name)
```

```
TABLESPACE users
```

```
STORAGE (INITIAL 20K NEXT 20k);
```


Lab 09 - Week 09

Topic: PL/SQL Basics

Task 1:

Write an PL/SQL program to add two numbers and the sum of those two numbers is stored in a third variable and print the result on the screen.

```
SQL> Declare
  2  -- declare variable x, y and z of datatype number
  3  x number(5);
  4  y number(5);
  5  z number(7);
  6  Begin
  7  -- Here we Assigning 10 into x
  8  x:=10;
  9  -- Assigning 20 into x
 10  y:=20;
 11  -- Assigning sum of x and y into z
 12  z:=x+y;
 13  --Print the Result
 14  dbms_output.put_line('Sum is '||z);
 15  End;
 16  /
Sum is 30
```

Lab 10 - Week 10

Topic: PL/SQL Basics

Task 1:

Write an PL/SQL procedure that calculate factorial of a number using a function?

```
CREATE OR REPLACE FUNCTION fact(x number)
RETURN number IS
    f number;
BEGIN
    IF x=0 THEN
        f := 1;
    ELSE
        f := x * fact(x-1);
    END IF;
    RETURN f;
END;
/

DECLARE
    num number;
    factorial number;
BEGIN
    num:= 6;
    factorial := fact(num);
    dbms_output.put_line(' Factorial ' || num || ' is ' || factorial);
END;
/
```

Lab11-Week 11

Topic: Users information using Data Dictionary Views

Task 1:

Write a PL/SQL command for listing all the tablespace quota?

```
Select * from DBA_TS_QUOTAS;
```

Task2:

Write a PL/SQL command for listing all profiles and assigned limit?

```
Select * from DBA_PROFILES ORDER BY PROFILE;
```

Lab 12 – WEEK 12

Topic: Managing and Creating User with Commands

TASK 1:

Write down the syntax for creating an oracle user with commands?

```
CREATE USER sidney  
  IDENTIFIED BY out_standing1  
  DEFAULT TABLESPACE example  
  QUOTA 10M ON example  
  TEMPORARY TABLESPACE temp  
  QUOTA 5M ON system  
  PROFILE app_user  
  PASSWORD EXPIRE;
```

```
CREATE USER global_user  
  IDENTIFIED GLOBALLY AS 'CN=analyst  
  DEFAULT TABLESPACE example  
  QUOTA 5M ON example;
```

TASK 2:

Write a SQL command that display all tablespace quotas?

Answer: SELECT * FROM DBA_TS_QUOTAS;

TASK 3:

Write a SQL command that display user information like name, profile and account status from database.

```
SELECT USERNAME, PROFILE, ACCOUNT_STATUS FROM DBA_USERS;
```

Lab 13- week 13

Topic: Pre-defined Roles & Assigning Roles for DBA

Task 1:

Write the SQL command for altering the role of database administrator in oracle database?

```
ALTER ROLE role_name {NOT IDENTIFIED |IDENTIFIED  
{By password | USING Package | EXTERNALLY | GLOBALLY}};
```

Task 2:

List down the pre-defined roles of database administrator in oracle database?

CONTENT

RESOURCE

DBA

EXP_FULL_DATABASE

IMP_FULL_DATABASE

DELETE_CATALOG_ROLE

EXECUTE_CATALOG_ROLE

SELECT_CATALOG_ROLE

Lab 14 – Week 14

Topic: SQL* Loader

Task 1:

Using SQL* Loader copy (load) data from the “Dept” table into the “Dept1” table without APPEND hint.

```
INSERT INTO Dept1
```

```
SELECT * FROM Dept;
```

```
COMMIT;    -- commit command is necessary to load data
```

Task 2:

Using SQL* Loader copy (load) data from the “Product” table into the “Product1” table with APPEND hint.

```
INSERT /*+ APPEND */ INTO Product1
```

```
SELECT * FROM Product;
```

```
COMMIT;    -- commit command is necessary to load data
```

Lab 15 (Solution)

Topics: Transaction Processing & Role of Transactions

Task 1:

Supposed you have the following table named “Employee”. Write a Transaction (PL/SQL code) to update the salary of ‘4156897’ employee number to 17900.

Employee		
EMP_No	EMP_Title	Salary
4156897	Talha Akhtar	15500
3256984	Ali Khalid	21500

Solution:

```
BEGIN
```

```
UPDATE Employee
```

```
SET Salary=17900
```

```
WHERE EMP_NO=4156897;
```

```
DBMS_OUTPUT_LINE(“Transaction updated successfully”)
```

```
END;
```

Lab 16 (Solution)

Topics: Transaction Processing & Role of Transactions

Task 1:

Write SQL command to allocate (grant) SELECT, INSERT and UPDATE privileges on “Credit” table to all users.

Solution:

```
GRANT      SELECT, INSERT, UPDATE
ON         Credit
TO         PUBLIC;
```

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